

JavaScript - Labo_07

Demo

Labo begint met het kijken van de Demo's die richten op de geziene leerstof tijdens de les. Het gaat over time intervals en dergelijke.

Hit an Object

Bij hit an object had ik bij men javascript meerder functies nodig namelijk, start, end en move. Dan ziet het er als volgt uit:

HTML:

```
<!DOCTYPE html>
<html lang="nl">
  <head>
    <meta charset="UTF-8" />
    <link rel="stylesheet" href="styles/style.css" />
    <title>Hit an Object</title>
  </head>
  <body>
    <button id="btnStart">Start</button>
    <p>
      <span>score: </span>
      <span id="score"></span>
    </p>
    <div id="playField">
      
    </div>

    <script src="script/code.js"></script>
  </body>
</html>
```

JavaScript:

```
let global = {
  IMAGE_COUNT: 5, // aantal figuren
```

```

IMAGE_SIZE: 48, // grootte van de figuur
IMAGE_PATH_PREFIX: "images/", // map van de figuren
IMAGE_PATH_SUFFIX: ".png", // extensie van de figuren

MOVE_DELAY: 3000, // aantal milliseconden voor een nieuwe afbeelding v
erschijnt

score: 0, // aantal hits
timeoutId: 0, // id van de timeout timer, zodat we die kunnen annuleren
};

const setup = () => {
  document.getElementById("btnStart").addEventListener("click", start);
};

const start = () => {
  clearInterval(global.timeoutId);
  document
    .getElementsByClassName("imgBomb")[0]
    .addEventListener("click", updateScore);
  global.timeoutId = setInterval(move, 1000);
  global.score = 0;
  score.innerText = global.score;
};

const end = () => {
  clearInterval(global.timeoutId);
  window.alert(`GAME OVER\nFinal Score = ${global.score}`);
};

const updateScore = () => {
  // Check of het een bom is of niet
  let image = document.getElementsByClassName("imgBomb")[0];
  if (image.getAttribute("src") === "images/0.png") {
    global.score -= 1;
    score.innerText = global.score;
    end();
  } else {
    global.score += 1;
    score.innerText = global.score;
  }
};

```

```

    }
  };
  const move = () => {
    let image = document.getElementsByClassName("imgBomb")[0];
    let playfield = document.getElementById("playField");
    let maxLeft = playfield.clientWidth - image.offsetWidth;
    let maxHeight = playfield.clientHeight - image.offsetHeight;

    // verplaats de sprite
    let left = Math.floor(Math.random() * maxLeft);
    let top = Math.floor(Math.random() * maxHeight);
    image.style.left = left + "px";
    image.style.top = top + "px";

    // randomize image
    let randomNumber = Math.floor(Math.random() * global.IMAGE_COUNT);
    image.setAttribute(
      "src",
      global.IMAGE_PATH_PREFIX + randomNumber + global.IMAGE_PATH_SUFFIX
    );
  };

  window.addEventListener("load", setup);

```

Matching game

HTML:

```

<!DOCTYPE html>
<html lang="nl">
  <head>
    <meta charset="UTF-8" />
    <link rel="stylesheet" href="styles/style.css" />
    <title>Matching game</title>
  </head>
  <body>
    <h2>Matching game</h2>

```

```

<button id="btnStart">Start</button>
<button id="btnReset">Reset</button>
<span>Score: </span><span id="score"></span>
<div id="cardContainer"></div>

<script src="scripts/code.js"></script>
</body>
</html>

```

JavaScript

```

let global = {
  AANTAL_HORIZONTAAL: 4,
  AANTAL_VERTICAAL: 3,
  AANTAL_KAARTEN: 6,
  IMAGE_PREFIX: "images/",
  IMAGE_SUFFIX: ".jpg",
  IMAGES: ["kaart1", "kaart2", "kaart3", "kaart4", "kaart5", "kaart6"],
  executed: 0,
  flippedCards: [],
  matchedPairs: 0,
};

const setup = () => {
  document.getElementById("btnStart").addEventListener("click", start);
  document.getElementById("btnReset").addEventListener("click", reset);
};

const start = () => {
  const cardContainer = document.getElementById("cardContainer");
  let cardImages = [...global.IMAGES, ...global.IMAGES];
  cardImages = shuffle(cardImages);

  const score = document.getElementById("score");
  score.innerText = `${global.matchedPairs}`;

  if (global.executed === 0) {

```

```

for (let i = 0; i < cardImages.length; i++) {
  const img = document.createElement("img");
  img.src = `${global.IMAGE_PREFIX}achterkant${global.IMAGE_SUFFIX}`;
  ;
  img.alt = `Card`;
  img.dataset.index = `${i}`;
  img.dataset.image = cardImages[i];
  img.addEventListener("click", turnCard);
  cardContainer.appendChild(img);

  global.executed = 1;
}
};

const reset = () => {
  const cardContainer = document.getElementById("cardContainer");
  cardContainer.innerHTML = ""; // Clear previous game
  global.executed = 0;
  global.matchedPairs = 0;
  global.flippedCards = [];
  const score = document.getElementById("score");
  score.innerText = `${global.matchedPairs}`;

  start();
};

const turnCard = (event) => {
  let clickedCard = event.target;

  if (
    global.flippedCards.length === 2 ||
    clickedCard.classList.contains("matched")
  )
    return;

  clickedCard.src = `${global.IMAGE_PREFIX}${clickedCard.dataset.image}${global.IMAGE_SUFFIX}`;
  global.flippedCards.push(clickedCard);

```

```

if (global.flippedCards.length === 2) {
  checkMatch();
}
};

const checkMatch = () => {
  let [card1, card2] = global.flippedCards;
  const score = document.getElementById("score");

  if (card1.dataset.image === card2.dataset.image) {
    // Match found
    card1.classList.add("matched");
    card2.classList.add("matched");
    global.matchedPairs++;
    score.innerText = global.matchedPairs;

    if (global.matchedPairs === global.AANTAL_KAARTEN) {
      setTimeout(() => alert("You win!"), 500);
    }
    global.flippedCards = [];
  } else {
    // No match, flip them back after a short delay
    setTimeout(() => {
      card1.src = `${global.IMAGE_PREFIX}achterkant${global.IMAGE_SUFFIX}`;
      card2.src = `${global.IMAGE_PREFIX}achterkant${global.IMAGE_SUFFIX}`;
      global.flippedCards = [];
    }, 1000);
  }
};

const shuffle = (array) => {
  return array.sort(() => Math.random() - 0.5);
};

window.addEventListener("load", setup);

```